

Figure 1: KGDB with the VirtualBox set-up

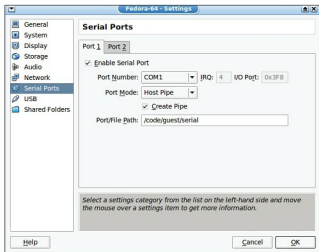


Figure 2: Configuring the serial port in VirtualBox

for the kernel source directory. This share is mounted within the guest, and the kernel is compiled and installed from the guest command prompt. View Figure 3 for an idea of my set-up.

There are many benefits in such a set-up—the shared kernel source can be used to directly do a ‘make install’ of the kernel within the guest. While debugging, you will need the kernel source files in the host OS, so that they are accessible to GDB—and let’s not forget the `vmlinux` file, which is passed as one of the arguments to the debugger. If you are debugging kernel modules, you can edit the module source (in the NFS-shared folder) from the host, while you debug the guest. There are many other ways of doing this, which you can explore on the Internet.

## Preparing and installing the kernel on a guest OS

The kernel source can be compiled either on the guest or the host. Since I have Ubuntu on the host and Fedora in the guest, I preferred to compile the kernel in the guest itself. Wherever you compile it, you will obviously need the build environment set up. Again, preparing a build environment is a task that is documented very well on the Internet. An important point to note is that compiling the kernel in the VM (guest OS) is extremely time-consuming.

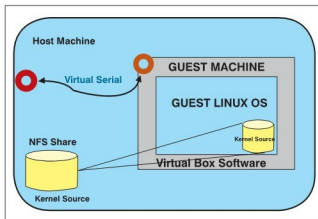


Figure 3: The NFS share set-up with VirtualBox

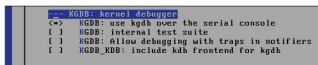


Figure 4: KGDB serial console options

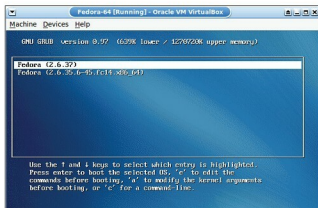


Figure 5: Newly compiled kernel in the GRUB menu

- During kernel configuration (once you do a ‘make menuconfig’) ensure you enable the following options:
- Kernel hacking-> (options for kernel hacking)
  - Kernel debugging (features for kernel debugging)
  - Compile the kernel with debug info (kernel and modules are compiled with the `-g` option)
  - Compile the kernel with frame pointers (frame-pointer registers are used to keep track of stack)
  - KGDB: Kernel debugger-> (enable KGDB)
  - KGDB: Use over serial console (enable serial console support) (see Figure 4)

Once the kernel is compiled, if you do a `make modules_install` and `install` from within the guest, it will install the newly compiled kernel. Figure 5 shows the Grub option for the new kernel after a guest reboot.

## Editing the boot-loader

To enable KGDB serial console support from the guest, we need to append the options `kgdboc=ty50` and `115200`